

Genetic Algorithm based Chain Leader Election in Wireless Sensor Network for Precision Farming

Hamzarul Alif Hamzah

Modelling, Simulation & Computing Laboratory
Faculty of Engineering, Universiti Malaysia Sabah
Kota Kinabalu, Malaysia.
hahamzah@ieee.org, msclab@ums.edu.my

Norah Tuah

Faculty of Computer and Mathematical Sciences
Universiti Teknologi MARA
Kota Kinabalu, Malaysia.
norahtuah@uitm.edu.my

Kit Guan Lim

Modelling, Simulation & Computing Laboratory
Faculty of Engineering, Universiti Malaysia Sabah
Kota Kinabalu, Malaysia.
kitguan.lim.my@ieee.org, msclab@ums.edu.my

Min Keng Tan

Modelling, Simulation & Computing Laboratory
Faculty of Engineering, Universiti Malaysia Sabah
Kota Kinabalu, Malaysia.
mktan@ieee.org, msclab@ums.edu.my

Ismail Saad

Faculty of Engineering
Universiti Malaysia Sabah
Kota Kinabalu, Malaysia.
ismail_s@ums.edu.my

Kenneth Tze Kin Teo*

Modelling, Simulation & Computing Laboratory
Faculty of Engineering, Universiti Malaysia Sabah
Kota Kinabalu, Malaysia.
kktteo@ieee.org, msclab@ums.edu.my

Abstract—Wireless Sensor Network (WSN) is one of the commonly used technologies in Precision Farming (PF). It provides farmers with accurate real-time information on their farms. In practice, WSN consists of numerous wireless sensor nodes, where each node relies on limited energy sources such as battery to maintain its operation. The energy management issue in WSN has gained attention of scholars, leading to new protocols or schemes developed over the years. Conventionally, PEGASIS protocol selects chain leader without considering the distance and residual energy level of each sensor node. As such, it might increase the energy consumption rate to transmit collected data from sensor node to sink. Inadequate energy management leads to rapid energy drain and eventually shorten the lifespan of WSN. Therefore, this study aims to prolong the lifespan of WSN while minimizing the energy consumption. Genetic Algorithm (GA) is proposed to enhance the chain leader selection scheme of the conventional PEGASIS. The proposed GA will select optimum chain leader by considering the energy consumption rate of each node. As such, the proposed algorithm is able to increase node's transmission as well as improve the lifespan of WSN by 50 % as compared to the conventional approach.

Keywords—genetic algorithm, power-efficient gathering in sensor information systems, wireless sensor network, precision farming

I. INTRODUCTION

Agricultural sector plays an important role in supporting the growth of global population, which is projected at 9.7 billion people by 2050. World Bank reported that the livelihood of 65 % low-income population depends on the agricultural sector [1]. In order to support the rapid population growth, new technologies have been employed to assist in farm management and optimizing productivity. Wireless Sensor Network (WSN) is one of the well-known methods employed in Precision Farming (PF) [2]. In most of the applications, WSN is implemented as a platform to provide farmers with real-time information on current conditions of their farm. With WSN technology, farmers can detect abnormalities in their farms, e.g. soil humidity and pH levels, and react accordingly to these abnormalities. As such, crops yield's quality and quantity can be enhanced.

Generally, WSN is deployed to continuously monitor the environmental changes in the farm. However, continuous

monitoring will rapidly drain energy of each sensor nodes within the network. The inefficient energy management poses a great challenge to the sustainability of WSN as farmers need to replace energy sources regularly [3]. Thus, optimizing the energy management of WSN will improve lifespan of sensor nodes as well as to minimize the maintenance costs. Power-Efficient Gathering in Sensor Information System (PEGASIS) protocol has been proposed to enhance the energy management of WSN. PEGASIS will determine data transmission path between nodes and base station [4]. PEGASIS is proven to be able to reduce energy consumption in contrast to the primitive direct transmission method [5].

Conventionally, PEGASIS selects chain leader based on rotating basis [6, 7]. Each node, without exception, will be selected as the chain leader by turns. Since the protocol does not consider residual energy of each sensor node and the distance between sensor nodes and sink, PEGASIS is inadequate to promote energy balance among the sensor nodes [8]. For instance, Node A, which is located 50 m away from the base station, has equal chance to be selected as Node B, which is located 5 m away from the base station. Since energy needed for transmitting a data is corresponding to the distance between source and the destination, Node A will require higher energy than Node B to transmit data, and eventually the energy of Node A will deplete faster. The imbalance is also manifested at the end of transmission rounds, where lifespan of the single node could be prolonged than its peers. Therefore, this study would like to propose Genetic Algorithm (GA) to prolong the lifespan of WSN by improving the chain leader selection mechanism of PEGASIS. The proposed algorithm will be able to balance the energy consumption of all sensor nodes within the network.

This paper is organized into few sections, where Section II presents the modelling of WSN which will act as the platform to evaluate robustness of the proposed algorithm. Section III presents the development of the proposed algorithm. Section IV discusses the result of the conventional and proposed algorithm under different test environments, and also the improvements that were made towards system performance enhancement, in the perspectives of network lifetime and individual transmission. Section V summarizes the findings of this work and proposes a recommendation for future study.

* corresponding author

This work is supported by UMS SDK, grant no. DN20086, UMSGreat, grant no. GUG0444-1/2020 and UMS Postgraduate Assistant Scheme (SBP).

II. REVIEW OF CLUSTER HEAD SELECTION APPROACHES

Energy management is a problem that has long impeded the WSN system. Many scholars are gearing on this issue to establish a WSN structure that could perform longer while at the same time maintains data transmission accuracy. Some scholars had developed routing protocols to act as the WSN's 'manager', which had helped tremendously to enhance the WSN's energy management [9]. With these protocols established, many consequence studies on WSN had becomes more specific. Scholars' attention has shifted to devise new techniques that can complement or improve these protocols. Among the most well-known techniques focused are path routing [3, 5] and cluster head election [10, 11]. For this study, election techniques for cluster head, or better known as chain leader in PEGASIS protocol is devised.

Cluster-based and chain-based protocol are amongst the most well-known protocols in WSN. The most distinct difference between the two is their method of data transmission. Cluster-based protocol separates its nodes into several clusters which are usually dependent on the parameters of location or residual energy level [5, 10]. A leader is elected in each cluster; hence its name is also known as cluster head. Cluster head will collect data from all nodes within its clusters and transmit gathered data to the sink. Chain-based, in the opposite, does not distribute its nodes into groups. Instead, it chains all nodes whereby the routing of data transmission will follow the order of this chain. The chaining process is similar to the Travelling Salesman Problem (TSP). Next, a chain leader will be selected among the nodes within this chain to gather data and send them to the sink.

There are a lot of approaches that can be taken to enhance the cluster head election scheme [12]. However, not much information can be found that provides a specific focus on chain leader election in PEGASIS protocol. Regardless, scholars mainly study and integrate parameters

of residual energy and distance to the sink in finding optimal cluster head irrespective of the protocol it is implemented in [10]. Some also use node density in the fitness function [11] but it is only suitable for cluster-based protocol and is unsuitable for PEGASIS, which is a chain-based protocol. Both parameters, residual energy and distance to the sink are studied for this research with GA as the base algorithm.

GA and other Artificial Intelligence (AI) algorithms are not foreign in WSN [13, 14]. Many researchers have been using AI algorithms in their study as it is robust, efficient and accurate in finding an optimal solution to a problem. Lee *et al.* implemented GA in VANET to solve the dynamic multicast problem and improve the efficiency of packet delivery [15]. Other common problems include clustering, path routing [3, 5] and cluster head election [11, 16, 17]. For instance, Altakhayneh *et al.* use GA for cluster head selection in LEACH, which is a cluster-based network with residual energy as its focus [10]. Similar recent research by Behera *et al.* (2019) also develops a GA-based cluster head selection that is based on residual energy [16]. However, it did not provide an emphasis on the distance to the sink parameter.

Another example of AI implementation in the cluster head selection scheme is by Alghamdi (2020) that considers four criteria; energy, delay, distance and security [18]. Alghamdi proposed a hybridized concept of dragonfly and firefly algorithms. Zhu and Zhang use PSO for path optimization and considers residual energy and distance between nodes as the fitness function [14]. Moreover, spider monkey optimization was also used for cluster-head selection by Lee *et al.* and Gui *et al.* [12, 13] to improve the lifetime and stability of WSN. While the use of AI is common, it does not mean that an AI algorithm can be chosen hastily. Ability and compatibility of an algorithm are crucial in obtaining accuracy in the implemented environment [19, 20]. The method of implementation is also a subject that needs to be given enough attention.

TABLE. I. SUMMARY OF CLUSTER HEAD SELECTION APPROACHES REVIEW

Parameters	Description	Algorithm	Literature
Residual energy	- Proposed two stages clustering algorithm; set-up and steady-state - Clusters and cluster heads are formed with the typical LEACH mechanism for the first round - Modified equation for next rounds, considers residual energy and an optimal cluster's number	LEACH	[16]
Distance to sink, residual energy, density	- Developed G-LEACH - Data transmission start after the best energy nodes is elected as cluster head using GA. - GA also used in the formation of clusters	GA	[10]
Residual energy, distance between nodes, distance to sink and workload	- ARSH-FATI-based Cluster Head Selection (ARSH-FATI-CHS) - Aimed to minimize the energy consumption and maximize network lifetime - Improves network lifetime by maximizing minimum lifetime in the network.	ARSH-FATI	[19]
Energy consumption, density	- Devised sampling-based Spider Monkey Optimization (SMO) - Highest weight for CH's distribution in a cluster as its impact is large - Consider reserve energy (energy consumed and left when a node become cluster head.	SMO	[12]

III. MODELLING OF WIRELESS SENSOR NETWORK

In this study, a WSN model with 100 sensor nodes is simulated in MATLAB. All simulated nodes are distributed randomly in a 100 m x 100 m area [21]. Base station or sink is located at the middle of the area with coordinate of (50, 50). This is illustrated in Fig. 1. This study also uses a similar radio model that was used in the study of Lindsey and Raghavendra [21]. Amount of energy used to operate the transmitter or receiver circuitry is called E_{elec} while ϵ_{amp} is the energy that the transmitter amplifier dissipates. Energy costs for the process of receiving and transmitting a k -bit message at a distance d are calculated using (1) to (6), where (3) and (4) are used for transmitting data and (5) and (6) are used for receiving data.

$$E_{elec} = 50 \times 10^{-9} \text{ J bit}^{-1} \quad (1)$$

$$\epsilon_{amp} = 100 \times 10^{-12} \text{ J bit}^{-1} \text{ m}^{-2} \quad (2)$$

For transmitter,

$$E_{Tx}(k, d) = E_{Tx-elec}(k) + E_{Tx-amp}(k, d) \quad (3)$$

$$E_{Tx}(k, d) = E_{elec} \times k + \epsilon_{amp} \times k \times d^2 \quad (4)$$

For receiver,

$$E_{Rx}(k) = E_{Rx-elec}(k) \quad (5)$$

$$E_{Rx}(k) = E_{elec} \times k \quad (6)$$

The k and d used in above equations are packet length and distance respectively. The packet length is set to 4,000 bits in the simulation. Additional energy is spent for the middle nodes which are data aggregation energy, E_{DA} . E_{DA} used in this study is fixed at 5 nJ/bit/message, as according to Lindsey and Raghavendra [21]. The equation is shown as follow:

For transmitter,

$$E_{Tx}(k, d) = (E_{elec} + E_{DA}) \times k + \epsilon_{amp} \times k \times d^2 \quad (7)$$

For receiver

$$E_{Rx}(k) = (E_{elec} + E_{DA}) \times k \quad (8)$$

For data transmission in PEGASIS, a chain connecting all nodes are required. Typical PEGASIS build its chain with greedy algorithm. The chain will start at the node furthest from the sink. Greedy mechanism will evaluate all nodes' location and find the nearest node to be designated as the 'next' node in the order of the chain. In next iteration, 'next' node is set as 'current' node whereby the process will repeat continuously to find the nearest node to be selected as the 'next' node. This procedure will repeat until all nodes are chained. The nodes that are already chained will not be selected again. A chain example in WSN is shown in Fig. 1.

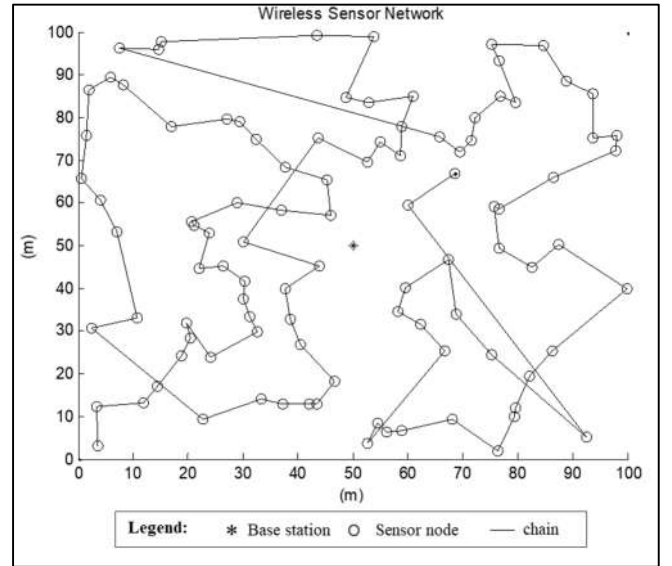


Fig. 1. Topology for 100 nodes and sink in 100 m x 100 m area.

Before the data transmission procedure in PEGASIS, a chain leader is elected. Conventional PEGASIS mechanism elects chain leader by turns, following the order of the chain. This means, first node in the chain order will be the first elected chain leader for the first round. Similarly, 100th node in the chain order is also the 100th round's chain leader. This method is improvised with a proposed procedure in Section IV. Then, for data transmission, the process will follow the chain arrangements. For instance, when Node 1 becomes the chain leader, data transmission starts from the last node where it operates multi-hop and data aggregation along the route to Node 1. When all data reached Node 1, it will send those received data to the base station. In the contrary, when a middle node is selected as the chain leader, data transmission will start from both ends until all data are gathered at the middle node. Then the middle node will transmit the collected data to the base station.

IV. GENETIC ALGORITHM ADAPTATION

The main objective of this study is to devise an alternative for the conventional election-by-turn to a scheme that is aware of residual energy level (E_{res}) and node's distance to sink (d_{sink}). GA is chosen as the algorithm to perform this task. It is noted that E_{res} will change after each round of transmission due to node's energy consumption, which includes transmitting energy and receiving energy. As distance between a node to another node in the chain is different as according to their location, energy spent for data transmission will differ respectively based on the distance, whereby longer distance will cost higher energy consumption and shorter distance incur lower transmitting energy. In the opposite, d_{sink} is constant throughout the network lifetime as nodes' position is fixed.

Since nodes spent different amount of energy for data transmission based on the distance size that it possesses, some nodes can be seen spending higher energy than another. To introduce a balance state in energy consumption, nodes that have a lower residual energy level needs to be avoided during chain leader election, while those with higher residual energy will be given a higher chance to be elected. Therefore, the fitness is directly proportional to E_{res} as in (9) where i is the chromosome in the population n .

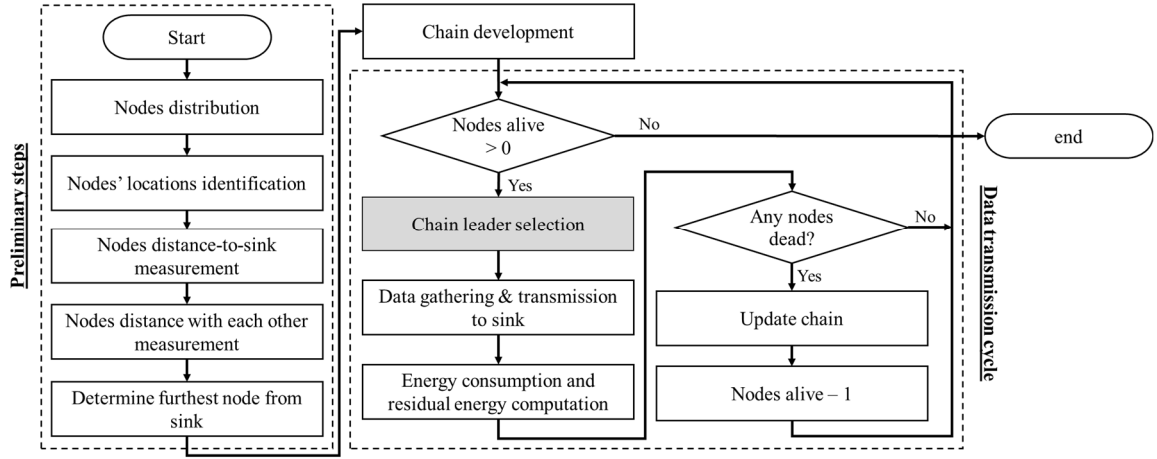


Fig. 2. Flowchart of GA based PEGASIS.

Next fitness function is to focus on the parameter of distance from a node to the sink. As shown in Fig. 1, nodes are distributed in an area with a sink in the middle. Therefore, distance of each node to the sink is different. Nodes that are nearby the sink will have shorter distance than those farther, thus will spend lower energy cost. Hence, in terms of d_{sink} , shorter distance should have higher fitness while longer distance should have lower fitness. The fitness is shown in (10).

$$f_{1,i} = E_{res_{-1,i}}, i = 1, 2, 3, \dots, n \quad (9)$$

$$f_{2,i} = \frac{1}{d_{sink_{2,i}}}, i = 1, 2, 3, \dots, n \quad (10)$$

To associate both fitness (9) and (10), weightage is introduced where weightage 1 (w_1) is for fitness 1 while weightage 2 (w_2) is for fitness 2. The value of both weights must be within 0 to 1, and the addition of w_1 and w_2 will be equal to 1. Thus, fitness used for GA in this study is shown in (11).

$$f = (w_1 \times f_{1,i}) + (w_2 \times f_{2,i}) \\ = (w_1 \times E_{res_{-1,i}}) + \left(w_2 \times \frac{1}{d_{sink_{2,i}}} \right) \quad (11)$$

where $w_1 + w_2 = 1$.

In GA, a solution space records the node ID of the chromosome, fitness, rank, probability and cumulative probability in columns 1 to 5 respectively. As the solution space for node's identification number (ID) and fitness of the chromosome is filled, it is then rearranged in increasing order of chromosomes' fitness where the rank is recorded in column 3. Next, probability of each chromosome is computed using (12) and lastly, column 5 is filled with cumulative probability using (13).

$$P_i = \frac{r_i}{\sum_{i=1}^n r} \quad (12)$$

$$CP_i = \sum_{i=1}^n (CP_i + P_i) \quad (13)$$

Fig. 2 shows the location of GA implementation within PEGASIS mechanisms. As explained previously, GA is adopted for chain leader selection which is a primary step in a cycle of data transmission, before proceeding to the data gathering and transmission to sink. The parameters being initialized in GA includes population size, generation, probability crossover, probability mutation and crossover value. These parameters need to be trained by performing multiple tests to obtain a compatible value set with PEGASIS.

Crossover probability determines the regularity of crossover being performed. When this parameter is set to 0.0, crossover will not occur and the offspring will be an exact copy of its parents. This is not ideal for this study nor most of the others since the offspring need to evolves to find the best chain leader or cluster head. In the opposite, when crossover probability is 1.0, crossover is performed in all generations, hence offspring will always differ from its parents. Best crossover practice, especially for this study is when offspring keeps the good parts of parents, so that the offspring will keep better values in each generation. Mutation on the other hand is crucial to prevent GA from falling into a local extreme. However, it should not operate too often as it will turn GA into a random search instead.

After crossover and mutation, it is possible that the offspring produced did not match any nodes in the solution pool. Hence, it needs to be repositioned so the offspring is always within the existing solution. To do so, the solution is replaced with nodes that contain the nearest fitness. Other than to ensure the accuracy of the result, this process also helps faster convergence of the population. For the weightage selection, a test is conducted to observe and analyze the impact of E_{res} and d_{sink} towards the performance of PEGASIS. Two results are observed which are the network lifetime and the rounds until the first dead node occurs. Network lifetime can give an insight into the performance of PEGASIS as a whole, while rounds of the first dead node can be used to analyze the ability of the proposed mechanism in providing balance system. The results are recorded in Table II.

TABLE. II. TEST ON WEIGHTAGE AND ITS IMPACT ON PEGASIS'S PERFORMANCE

Weightage Set	1	2	3	4	5	6	7	8	9
w1	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
w2	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
Network Lifetime	4572	4464	4410	4365	4342	4329	4320	4313	4309
	Lifespan showed a decreasing trend with higher dependency on E_{res} and lower dependency on d_{sink}								
Round of First Dead Node	976	976	1274	1435	1486	1499	1503	1502	1498
	First dead node's lifespan shown an increasing trend with higher dependency on E_{res} and lower dependency on d_{sink}								

Results in Table II shows that a higher dependency on E_{res} will cause the network lifetime to decrease. When energy is prioritized, location of nodes becomes less significant in the fitness, hence nodes located far away from nodes and possess high energy, will have the same chance as node located nearby the sink with a similar energy level, to be elected as next node. This is illustrated in Table III. Percentage difference of fitness of Node 2 which has a longer distance but similar E_{res} with Node 1, shown a decreasing trend difference with as low as 0.1691 %. It is possible that Node 2 is elected when GA algorithm is trapped in a local extremum.

In the opposite, a higher reliance on d_{sink} causes the first node to die at an earlier round of transmission. Most of the chain leaders elected are the ones located nearby the sink regardless of its residual energy. Nearby nodes that are burdened with high node-to-node distance will be affected heavily if they keep being selected as the chain leader. Therefore, both parameters are important and could affect PEGASIS performance as a whole. From Table III, weightage set 5 is seen to be more reliable with a higher lifetime and longer round of first dead node.

V. RESULTS AND DISCUSSION

Performance of the developed GA-based chain leader election mechanism is evaluated mainly in terms of total network lifetime and the round of first dead node. Besides, the extension of each node's lifetime is also studied by analyzing the total transmission of each node.

A. Network Lifetime and first dead node's round

It is common for PEGASIS with a conventional chain-election mechanism to have a longer network lifetime than GA-based mechanism. Nevertheless, the extension is mostly due to the end rounds of transmission as shown in Fig. 3. Nodes with conventional mechanism deplete its energy at a faster rate than GA, as recorded within the transmission round of 4,000 to 4,300. A node can be seen to perform the last rounds of transmission by itself, depleting its energy only by transmitting data to the sink. The length of the final lifespan is proportional to the residual energy of the node. Longer lifespan shows that the residual energy it possesses is higher, which also shows an imbalance of energy consumption between nodes in the earlier rounds. Despite having a longer lifespan, the fact that only several nodes are surviving would not be helpful to the users especially farmers in assisting their monitoring ability. GA-based mechanism replaces this imbalance by promoting fairer usage among nodes, at the same time distributing a balanced burden on the nodes by considering the E_{res} and d_{sink} .

TABLE. III. EXAMPLE FITNESSES EVALUATION

Node	E_{res}	d_{sink}	0.1	0.3	0.5	0.7	0.9
			0.9	0.7	0.5	0.3	0.1
1	1.5	10	0.2400	0.5200	0.8000	1.0800	1.3600
2	1.5	15	0.2100	0.4977	0.7833	1.0700	1.3577
% difference			12.5000	4.4885	2.0875	0.9259	0.1691
A decreasing trend with lower reliance on d_{sink}							

TABLE. IV. STANDARD DEVIATION OBSERVATION

Transmission Round	PEG_{conv}	GA	diff	% diff
500	0.0981	0.0886	0.0095	9.6992
1000	0.1946	0.1642	0.0305	15.6493
1500	0.2240	0.2437	-0.0197	-8.8118
2000	0.2378	0.2263	0.0114	4.7981
2500	0.2256	0.2119	0.0137	6.0745
3000	0.2140	0.1965	0.0175	8.1762
3500	0.1662	0.1432	0.0231	13.8778
4000	0.1148	0.0669	0.0479	41.7378

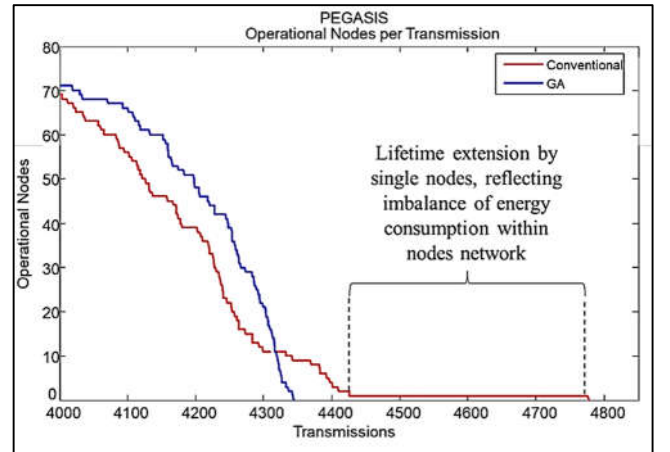


Fig. 3. Performance of Node's Lifetime under GA and Conventional.

Table IV shows the comparison on standard deviation records of nodes in different transmission round for conventional PEGASIS (PEG_{conv}) and GA mechanism. This record helps in examining the spread of energy consumption within the nodes in the network. Data is obtained by retrieving the residual energy of each node at the particular transmission rounds, and the standard deviation is calculated. GA has a more stable energy usage in most of the rounds, with as much as 42.38 % difference than the conventional method. This verifies the ability of GA mechanism in promoting fairer energy consumption within the node network.

TABLE V. EXTENDED NODE AND LIFETIME OBSERVATION

Observations	Results
Number of nodes with extended lifetime	67 nodes
Number of nodes with similar lifetime	1 node
Number of nodes with shortened lifetime	32 nodes
Highest extended individual transmission	944 transmissions
Total extended individual transmission	7,830 transmissions
Total shortened individual transmission	4,277 transmissions
Net extended individual transmission	3,553 transmissions

B. Individual Transmissions of Nodes

Table V summarized the results of several observations on the individual node's transmissions. With GA mechanism, 67 nodes have their life extended with the highest extension at 944 individual transmissions. One node recorded a similar number of transmissions with the conventional chain election technique while another 32 nodes showed shortened lifetime. The total number of extended transmissions is recorded at 7,830 transmissions and the total number of shortened-nodes' transmissions is recorded at 4,277 transmissions. Therefore, the extended transmissions are 3,553 transmissions. This further verifies the ability of GA in assisting balanced energy consumption within the node network and extending their lifespan.

VI. CONCLUSION

The performance and analysis of GA-based chain leader election have shown a positive outcome against the conventional election-by-turn mechanism. A proper fitness, focusing on the correct and crucial parameters are necessary for GA to perform well in the implemented platform. Moreover, this study also verifies the importance of energy balance in WSN as it will help to extend its nodes' individual lifespan.

ACKNOWLEDGEMENT

The authors would like to acknowledge Universiti Malaysia Sabah (UMS) for supporting this research under Niche Research Fund Scheme (SDN), grant no. DN20086, Postgraduate Research Grant Scheme (UMSGreat), grant no. GUG0444-1/2020, and Postgraduate Assistance Scheme (SBP).

REFERENCES

- [1] F. Preneuf, "Agriculture and Food," The World Bank, 2020.
- [2] F. Ouyang, H. Cheng, Y. Lan, Y. Zhang, X. Yin, J. Hu, X. Peng, G. Wang, and S. Chen, "Automatic Delivery and Recovery System of Wireless Sensor Networks (WSN) Nodes based on UAV for Agricultural Applications," *Computers and Electronics in Agriculture*, vol. 162, pp. 31 - 43, 2019, doi: 10.1016/j.compag.2019.03.025.
- [3] K.R. Khoubati, S.A. Solangi, Z. Bhatti, and D.N. Hakro, "Optimal Route Planning by Genetic Algorithm for Wireless Sensor Network," *Proceedings of International Conference on Information Science and Communication Technology*, 2020, doi: 10.1109/ICISCT49550.2020.9079944.
- [4] N. Ramluckun, and V. Bassoo, "Energy-Efficient Chain-Cluster Based Intelligent Routing Technique for Wireless Sensor Networks," *Applied Computing and Informatics*, 2018, doi: 10.1016/j.aci.2018.02.004.
- [5] J. Bhola, S. Soni, and G.K. Cheema, "Genetic Algorithm Based Optimized Leach Protocol for Energy Efficient Wireless Sensor Network," *Journal of Ambient Intelligence and Humanoid Computing*, 2019, doi: 10.1007/s12652-019-01382-3.
- [6] C.H. Lee, K.G. Lim, M.K. Tan, R.K.Y. Chin, and K.T.K. Teo, "A Genetic Algorithm for Management of Coding Resources in VANET," *Proceedings of IEEE 2nd International Conference on Automatic Control and Intelligent Systems*, 2017, pp. 80-85, doi: 10.1109/I2CACIS.2017.8239037.
- [7] A. Somauro and V. Bassoo, "Energy-Efficient Genetic Algorithm Variants of PEGASIS for 3D Wireless Sensor Networks," *Applied Computing and Informatics*, 2019, doi: 10.1016/j.aci.2019.07.002.
- [8] F. Sabery, and A. Hair, "Evaluation and Comparative Study of the Both Algorithm LEACH and PEGASIS Based on Energy Consumption," *Proceedings of 3rd International Conference on Networking, Information Systems & Security*, 2020, doi: 10.1145/3386723.3387838.
- [9] V. Chung, N. Tuah, K.G. Lim, M.K. Tan, H. Huang, and K.T.K. Teo, "Impact of Genetic Operators on Energy-Efficient Wireless Sensor Network," *Proceedings of IEEE 6th International Conference on Engineering Technologies and Applied Sciences*, 2019, pp. 1-6, doi: 10.1109/ICETAS48360.2019.9117419.
- [10] W.A. Altakhayneh, M. Ismail, M.A. Altahrawi, and M.K. AbuFoul, "Cluster Head Selection using Genetic Algorithm in Wireless Network," *Proceedings of IEEE 14th Malaysia International Conference on Communication*, 2019, doi: 10.1109/MICC48337.2019.9037609.
- [11] S. Verma, N. Sood, and A. K. Sharma, "Genetic Algorithm-Based Optimized Cluster Head Selection for Single and Multiple Data Sinks in Heterogeneous Wireless Sensor Network," *Applied Soft Computing Journal*, vol. 85, 2019 doi: 10.1016/j.asoc.2019.105788.
- [12] J.G. Lee, S. Chim, and H.H. Park, "Energy-Efficient Cluster-Head Selection for Wireless Sensor Networks using Sampling-Based Spider Monkey Optimization," *Sensors*, vol. 19, 2019, doi: 10.3390/s19235281.
- [13] T. Gui, F. Wang, C. Ma, and D.E. Wilkins, "On Cluster Head Selection in Monkey-Inspired Optimization Based Routing Protocol for WSNs," *International Conference on Computing, Networking, and Communication*, 2019, doi: 10.1109/ICCNC.2019.8685531.
- [14] X. Zhu, and Y. Zhang, "Wireless Sensor Network Path Optimization Based on Particle Swarm Algorithm," *Proceedings of IEEE International Conference on Computer Science and Automation Engineering*, 2011, doi: 10.1109/CSAE.2011.5952735.
- [15] J. Lee, J. Hwang, H. Park, and D. Kim, "On Latency-Aware Tree Topology Construction for Emergency Responding VANET Applications," *Proceedings of IEEE Conference on Computer Communications Workshops*, 2018, pp. 57-63, doi: 10.1109/INFCOMW.2018.8406819.
- [16] T.M. Behera, S.K. Mohapatra, U.C. Samal, M.S. Khan, M. Daneshmand, and A.H. Gandomi, "Residual Energy Based Cluster-Head Selection in WSNs for IoT Application," *IEEE Internet of things Journal*, 2019, doi: 10.1109/JIOT.2019.2897119.
- [17] J. Kim, Y. Kim, and Y. Yoo, "Energy-Efficient Routing Using Genetic Algorithm in Cluster-Based Wireless Sensor Networks," *Proceedings of IEEE 90th Vehicular Technology Conference*, 2019, doi: 10.1109/VTCFall.2019.8891489.
- [18] T.A. Alghamdi, "Energy Efficient Protocol in Wireless Sensor Network: Optimized Cluster Head Selection Model," *Telecommunication Systems*, vol. 74, pp. 331-345, 2020, doi: 10.1007/s11235-020-00659-9.
- [19] H. Ali, U. Tariq, M. Hussain, L. Lu, J. Panneerselvam, and X. Zhai, "ARSH-FATI a Novel Metaheuristic for Cluster Head Selection in Wireless Sensor Networks," *IEEE Systems Journal*, 2020, doi: 10.1109/JSYST.2020.2986811.
- [20] Z.W. Siew, Y.K. Chin, A. Kiring, H.P. Yoong, and K.T.K. Teo, "Comparative Study of Various Cluster Formation Algorithms in Wireless Sensor Networks," *Proceedings of 7th International Conference on Computing and Convergence Technology*, 2012, pp. 772-777.
- [21] S. Lindsey, and C. Raghavendra, "PEGASIS: Power-Efficient Gathering in Sensor Information Systems," *Proceedings of IEEE Aerospace Conference*, 2002, doi: 10.1109/AERO.2002.1035242.